

5.2 Malware

Malware (or malicious software) refers to any software that is designed to disrupt or damage a computer system or gain unauthorised access. Malware is used to steal personal data or financial reward.

There are various types of malware. These include:

- virus
- trojan
- worm
- spyware
- adware
- rootkit
- malicious bots
- ransomware.

It is important to understand how each type of malware operates and how to minimise the risk from each.

The consequences to the individual or to a business will depend on the type of malware. Some malware may simply cause a nuisance, such as slowing down a computer. Other types of malware may have more serious consequences, for example making data inaccessible or theft of personal data. This can lead to identity theft, corruption of data or blackmail.

Virus

A virus is a common type of malware and is the only type of malware that infects existing files on a computer system. The virus attaches itself to an uninfected file and modifies (changes) it, which means the file is now infected. The virus then replicates (copies) itself and attaches to another uninfected file. This process is repeated, with computer viruses designed to spread like a human virus.

Once a virus has infected a file, it may begin to replicate immediately, or it can lay inactive until actions performed by the computer cause the virus to start working. If an infected computer is part of a network, it may also begin to infect other computers on the network. A virus can be especially dangerous if it infects files on a server that are accessed by many different computers.

There are several signs that a computer has been infected with a virus, including:

- slower system performance
- files multiplying or duplicating automatically
- files being deleted without the user's knowledge.

The aim of a virus is to corrupt and disrupt data in a computer system.

DID YOU KNOW?

The first virus to infect PCs, named 'Brian', was created in 1986 and intended as an anti-piracy measure to stop people copying software. The virus was transferred via copied floppy disks (see Figure 5.7) and, instead of an infected machine starting normally, it displayed a message asking the user to call the virus's author to discuss how to obtain the software legally.



Figure 5.7: Floppy disks were used to store files and programs. Computers used to have floppy disk drives to read these, which worked in a similar way to CD, DVD and Blu-ray drives.

Preventing viruses

If a virus infects a computer system, it can be very difficult to remove. Anti-virus software should be installed on a computer to prevent the risk of infection by a virus. The computer should be regularly scanned for viruses. Most modern anti-virus software runs in the background scanning the computer for viruses while it is switched on. If the anti-virus software doesn't scan automatically, it is best to set up a daily automatic scan of the computer at a time when you know that the computer is likely to be switched on.

Any files the anti-virus software believes contain a virus will be **quarantined** and the user alerted. The user can then select to delete these files or investigate further. Alternatively, the anti-virus software can be configured to automatically delete all quarantined files.

The anti-virus software can detect a virus in a file by comparing the code to a database of known virus codes. If it finds a code that matches it will quarantine the file.

One weakness of anti-virus software is that it is dependent on the database it holds. If a system is infected with a virus not in the database, it will not be recognised and removed. The virus could cause significant damage, so it is important to regularly update anti-virus software to ensure it includes the latest known viruses.

Some anti-virus software can be set for real-time checking. This means that all programs that are requested for download are immediately checked for viruses. If the anti-virus software detects a virus, it will alert the user and tell them not to download the file, as it contains a virus.

A firewall can be used to minimise the impact and likelihood of being infected by a virus through filtering and monitoring incoming and outgoing traffic from a computer system. If the firewall detects malicious software trying to enter the system, it can stop it before it is able to enter. As with anti-virus software, a firewall again relies on the rules and filters to be kept up to date to detect the malicious traffic in the first place.

Care should also be taken when sharing resources using portable storage devices, such as a USB memory stick or external hard drive. USB memory sticks inserted into a computer should be scanned immediately, even if it is an individual's own storage device, as it could have been unknowingly infected by another machine. Some organisations do not allow the use of employees' personal memory sticks in work computers. This reduces the risk of viruses entering the network.

UNPLUGGED ACTIVITY 5.05

Discuss with your teacher what measures your school has in place to make sure that a virus is not installed on the school's computers. Write down any measures discussed that could also be used on your own device, or those used by your family.

Trojan

A trojan is a type of malware that disguises itself as a genuine file or software. A trojan can also be contained in software that may have been infected. A trojan is typically downloaded from an infected email or website. It hides within a malicious program. When the program is opened, it will often release another type of malware such as a virus, or attempt to steal personal data.

DID YOU KNOW?

The term 'trojan' comes from Greek mythology. There is a legend that the Ancient Greeks gave a large wooden horse as a peace offering to the people of the city of Troy (see Figure 5.8). Inside the 'trojan' horse, several Greek soldiers were hiding. When night fell, the soldiers climbed out of the horse and opened the city gates to let in more soldiers, who were then able to conquer the city of Troy.



Figure 5.8: What the Trojan horse might have looked like.

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Preventing trojans

It can be difficult to prevent trojans, as they disguise themselves as genuine software, which is unexpectedly run by the user. Anti-virus software can typically check for all types of malware, including trojans. Dedicated malware removal tools are also available, which could be used to remove a specific malware infection. In addition to using a firewall, the main way to reduce the risk of a trojan is to only open files and run software that you know is from a trusted source, such as from reputable software companies and trusted websites.

Questions

- 2 Find out what is meant by a backdoor trojan.

UNPLUGGED ACTIVITY 5.06

Imagine that your friend does not have any security installed on their computer. They think that no one will bother to hack into their computer. They do not realise that they could download a virus at any time. You need to convince your friend that they should take action to protect their computer. You could also advise them on what security measures they could take to prevent a virus being installed.

Write a one-minute speech to convince your friend. In pairs, practise the speech. Give your partner feedback. Say one positive thing about their speech and one thing that could be improved. Listen to your partner's feedback and improve your speech.

Worms

A worm is a type of malware that acts in a similar way to a virus, but with a major difference. A worm is a program that replicates itself, like a virus. But unlike a virus it does not need to attach itself to another program or file to cause damage. Worms can enter through security holes and issues in a computer, such as those within an operating system.

A worm replicates itself and typically aims to fill up all the free space on a computer to slow it down and stop it working (see Figure 5.9). Usually, the first sign for a user that their computer has been infected with a worm is the computer starts to run slowly and space on the hard drive begins to rapidly decrease.

A worm usually also tries to spread to other computers on a network. If a worm is able to spread throughout a network, it can block up bandwidth and slow down the whole network.

Worms are normally downloaded and spread through email attachments, peer-to-peer file-sharing networks or using a link to a website. Once downloaded, they will replicate.

A well-known example of a computer worm is Stuxnet, which was used as a cyber weapon in 2010. It was discovered by two security researchers who recognised that the worm was a great deal more complex than any they had previously seen. The worm was being used to attack a power plant and security experts believed the aim was to sabotage (stop) the production of nuclear weapons.

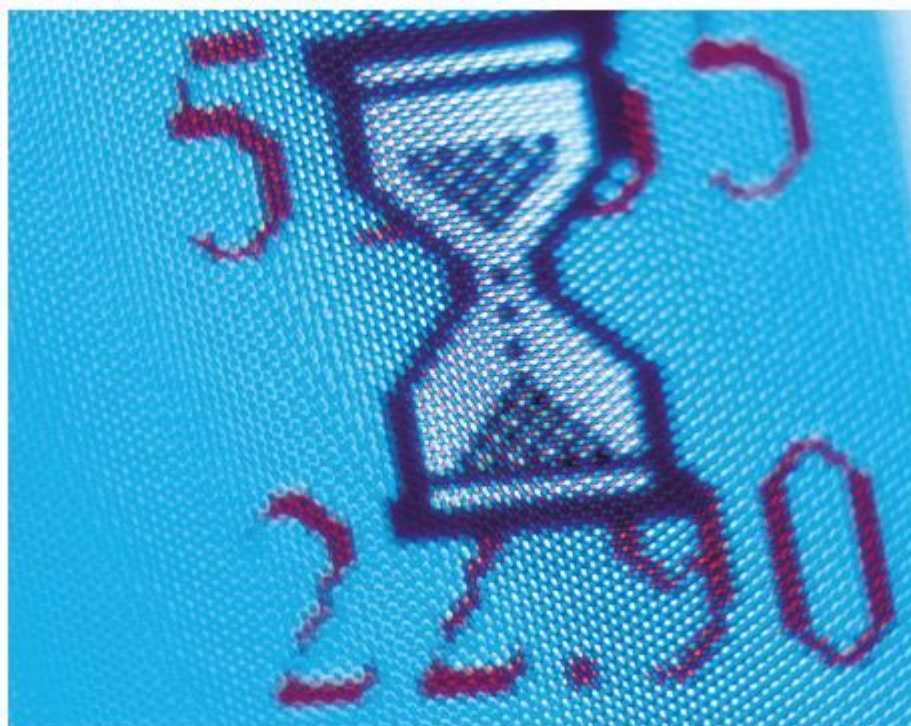


Figure 5.9: A worm can slow down the entire network.

Preventing worms

Worms often enter a computer through security weaknesses in the operating system or software installed by the user. It is important to regularly update the operating system and software to prevent worms. The update process can often be set to take place automatically, which removes the need for the task to be completed by the user.

Like phishing, a worm may be sent using an email, either as an attachment or through a hyperlink. Unless the email is from a trusted source, never open an attachment or click the

link. Instead, delete the email to reduce the risk of the computer being infected with a worm.

Anti-virus software usually checks for worms and, as they can be spread throughout a network, disconnecting a computer from a network when the network's resources are not required can help keep it safe. However, this may be impractical, as computers increasingly rely on a network connection, including to the internet, to use applications.

Spyware

Spyware describes malware designed to gather information about human interactions with a computer, such as key presses. As the name suggests, the aim of spyware is to spy on the user. Spyware is normally used to gather personal and sensitive data that can be used in fraudulent and criminal activity (see Figure 5.10).

A common example of spyware is a key logger. A key logger is installed on a user's computer, normally without the user knowing. The key logger will then record any key presses that are carried out by the user. All this data is sent to a third party, such as the person or group who created the spyware, to be analysed to establish any patterns in the data. The patterns are then examined to see if any of them look as though they could be personal or sensitive data, such as a password.



Figure 5.10: Spyware may gather information about you.

However, spyware can perform actions beyond key logging, such as:

- obtaining specific data within a system, such as contact lists or email addresses
- monitoring emails sent or websites visited
- obtaining audio, photos and videos taken on the device.

Spyware is often unknowingly downloaded while using the internet. It can be embedded into pop-up adverts offering a prize or a free product which entices the user to click them. It can also be embedded into the download of a video, music or application file, particularly if they are illegally made copies. Sometimes, this may involve the user unknowingly given consent to the spyware to be downloaded, as the agreement can go unnoticed within the terms and conditions.

Preventing spyware

As with other types of malware, it is important to check what is downloaded, especially when downloading software that is free of cost, or obtaining files from video and music sharing sites. Downloads should only be obtained from reputable (trustworthy) sources. In addition, links or offers in pop-up adverts should not be visited, no matter how enticing they may seem.

Terms and conditions (T&Cs) should be checked, along with user agreements, to ensure spyware is not included as part of software being installed. T&Cs could contain clauses about sharing data with third parties, which could also be obtained by spyware.

Anti-virus software can be used to scan a computer to remove spyware. If a key logger is found, it is advisable to change all passwords immediately, as this personal data could have been stolen.

ACTIVITY 5.07

- 1 How has the UK General Data Protection Regulation (GDPR) affected the use of spyware?
- 2 Find out about mobile spyware.
 - a How are mobile phones infected?
 - b How is spyware used in mobile devices?

Adware

Adware (advertising supported software) is a type of software that displays advertisements and pop-ups on a computer's browser. Adware collects data about a user's browsing habits. It then uses this information to customise advertisements targeted at the user's buying patterns and interests (see Figure 5.11).

Adware may be installed on a user's computer when software is downloaded. It may be present in the download without the user knowing. A user may only realise adware has been installed when they notice an additional task running on their device or search bar in their browser, or pop-ups start appearing. Adware may not be easy to delete, as there may be no uninstall option.

Some businesses use adware as a method of advertising, or to prompt users to use their search function. Adware used in this way may raise revenue for the business that, they argue, keeps down the cost of their products. Some users are happy to receive this type of targeted advertising. Adware can also be used maliciously causing the browser to keep displaying unwanted pop-ups and banners.



Figure 5.11: Adware uses browsing data to target advertising at individual customers.

Preventing adware

The most effective measure to avoid being infected with adware is to check exactly what is being downloaded and installed onto a computer. This could include examining the list of components that are included in the download and ensuring that any parts that could be adware, such as a search bar or task bar addition to the browser, are not installed.

As malware often does not have an uninstall option, it can be challenging to detect and remove, so a dedicated malware removal tool may need to be used. When a computer has multiple users, or forms part of a network, the system administrator may be the only one allowed to install software. This should help to prevent additional components being accidentally installed by non-specialist users.

DISCUSSION POINT

Should organisations restrict users' access to certain software or tasks on the organisation's computers, such as downloading software or using USB memory

sticks? Or is it more beneficial to train users on appropriate behaviour when using digital devices and accessing online services?

Rootkit

A rootkit is a computer program that enables an unauthorised person to gain administrator access to a victim's computer. A rootkit is designed to stay hidden on a user's computer, allowing the computer to be controlled by the unauthorised person from a remote location. A rootkit allows the unauthorised person to carry out criminal acts with the computer, such as hiding illegal files on the computer, using the computer as part of a larger cyber attack or to steal personal data (see Figure 5.12).

Rootkits can get installed because the user's device password is cracked or the security of a computer system is weak. The person controlling the rootkit can then use the access to stop the computer recognising that the rootkit is installed, so the victim will not know that someone else has complete access to their computer system. The rootkit will normally be hidden within the operating system, which makes it difficult to detect by anti-virus software. Other malware can be incorporated into a rootkit, which may be concealed on the computer to cause harm.

A rootkit may be installed on a computer by an individual purposely leaving USB memory sticks, infected with a rootkit, in places they believe they will be found. The individuals are relying on the curiosity of another user to insert the USB memory stick into their computer and therefore inadvertently install the rootkit.

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DID YOU KNOW?

An early rootkit example is NTRootkit, which appeared in 1999. It was designed to access Windows operating systems. A rootkit that was designed to do the same with Mac operating systems didn't appear until 10 years later.

ACTIVITY 5.08

Rootkits can be used a positive way. Research how rootkits are used in connection with Digital Rights Management (DRM). Make notes of your findings.



Figure 5.12: Never insert a USB memory stick into a computer unless you know it comes from a reliable source: it may be infected with Rootkit.

Preventing a rootkit

Here are some measures you can take to reduce the risk of infection by a rootkit:

- Set a strong password on your device to reduce the risk of it being cracked by an unauthorised person. Change the password regularly in case it has been discovered.
- Only download software from reputable sources.
- Never insert a USB memory stick into a computer system unless you trust the source.
- Removing a rootkit from a computer once it has been installed can be extremely difficult. It is software that is designed to be hidden and heavily relies on this feature. A rootkit may be detected by some anti-virus software or malware removal tools, although it may be necessary to completely reinstall the operating system.

Malicious bots

Bots (short for robots) are automated applications. A bot may be used to carry out simple, repetitive tasks that humans might find dull and time consuming, such as indexing a search engine. Malicious (intended to harm) bots are used by cybercriminals. This type of malware includes:

- Spam bots, which are used to bombard people's email inbox with spam (unwanted) emails, including those used to carry out other types of malware attacks.
- Zombie bots, which are used to create a bot network. The bot will lay hidden on a computer until an attack is launched. The computer will then connect to lots of other computers that have been infected with zombie bots to launch a large-scale attack on an organisation, such as a Distributed Denial of Service (DDoS) attack. Bots connected in this way are often known as a botnet.

- Malicious chatter bots, which pretend to be humans on certain websites or online services, such as social media platforms. They try to mimic human interaction, with the goal of obtaining personal data.
- File-sharing bots, which provide a fake link for a user's online search request. When clicked, the link downloads malware on to the user's computer.
- Malicious bots, which are like worms, as they can replicate and spread themselves.

DID YOU KNOW?

Large DDoS attacks can see networks hit with more than 300 million requests per second using infected machines across the globe, which makes computers and online services unusable.

Preventing malicious bots

Malicious bots are often embedded into links or software downloads. They are often spread in the same way as phishing attacks. So, the same measures used to protect against phishing can be applied to bots. Users should avoid clicking on any suspicious or unknown links, particularly if they are not from a reputable source.

As bots are often used by chat or messaging services, users should be careful when sharing personal data with a chat bot. Bots can even be used to mimic conversations with a user's contacts, such as family or friends, so individuals should still be careful about sharing personal data requested by others, particularly if the request is unexpected, or within a service not typically used.

If a user suspects they have downloaded a bot, anti-virus software can be used to detect and remove it. A firewall can also be used to detect the activity of malicious bots as it may recognise suspicious email traffic created by the bot. However, more sophisticated bots can disguise their communications as genuine network communications, so will go unnoticed by a firewall.

Questions

- 3 a State **one** way a bot can be used positively.
b State **two** ways a bot can be used maliciously.

Ransomware

Ransomware is a type of malware that restricts a user's access to their computer system and files. The ransomware will normally demand that the user pays a ransom, usually by transferring money so as to regain access to their computer system (see Figure 5.13). Some ransomware programs will completely lock a user's system, while others will encrypt all files on a system in a way that makes them useless. Ransomware will often enter a system in a similar way to a trojan. If the user doesn't pay the ransom in a set amount of time, they risk losing their data forever.

The likelihood of a person paying a ransom relies on the importance of the data to them, along with whether uninfected backups are available. Users that do pay the ransom can

often be exploited further and asked for additional payment to release their data.

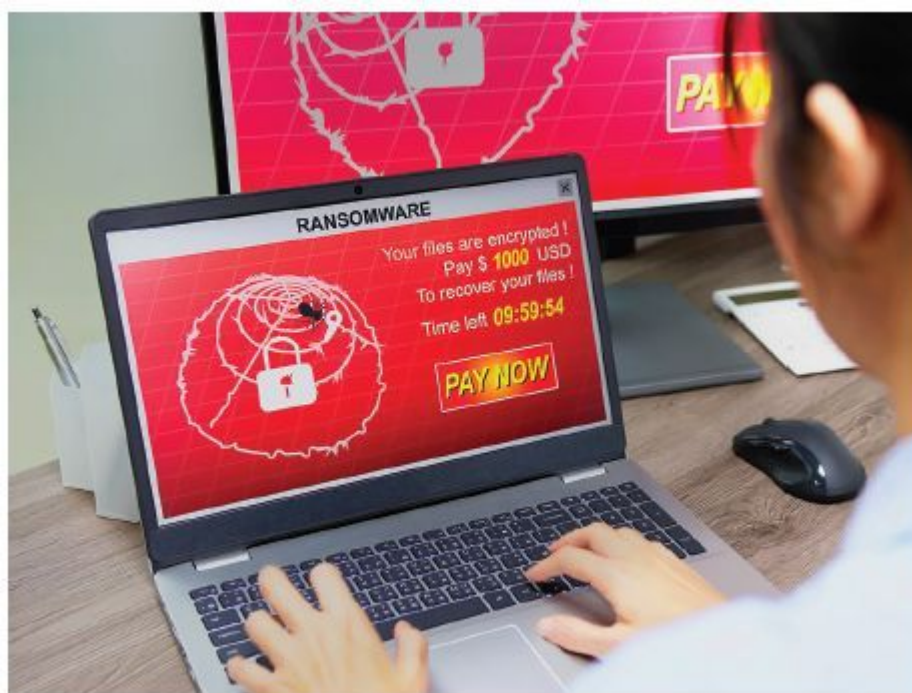


Figure 5.13: A Ransomware demand.

DID YOU KNOW?

A well-known example of Ransomware is WannaCry. The unauthorised person demands that the infected user pays a ransom. If the user does not pay within a certain period of time, the amount is increased. If payment is not made shortly after the amount is increased, the unauthorised user will delete the infected user's files forever.

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Preventing ransomware

As ransomware is distributed in a similar way to trojans, the same measures used to prevent trojans can be followed. Users can also limit the risk of losing data in this type of attack by making sure they have a copy of their data. If a separate back-up is available, which is kept up to date, then the risk of losing data in a ransomware attack becomes greatly reduced. This means users have a copy of their data if it is deleted during an attack.

The risk of losing data during a ransomware attack can also be reduced by storing data remotely in the cloud. Cloud servers often have a facility that allows the user to roll back to a previous version of the data, so it can be restored if lost.

ACTIVITY 5.09

An organisation can use a practice called principle of least privilege (POLP) to help keep data safe and secure. Research POLP and make a list of ways it can help keep data safe and secure. Share your list with a partner, and add more ways to your list.

Uses of malware

Malware is often used to carry out a number of illegal activities. These include:

- fraud
- theft
- industrial espionage
- sabotage.

Fraud

Fraud involves deceiving an individual or organisation into thinking a person is someone they are not. The aim of fraud is to obtain money or property. Malware can be used to make a user believe they are interacting with a reputable organisation when visiting their website. Instead, they are being deceived into using a fake website which is being used to gather personal data. The user may also be deceived into purchasing inferior quality products or services.

Personal data gathered through malware can be used to steal a person's identity. The theft of online banking details or credit cards may be used to purchase goods or apply for financial products, such as loans or benefits. Personal information can also be used to illegally obtain identification documents, such as passports or driving licences, which can be used to deceive others, for example to give false information when fined for a traffic offence.

Theft

As malware is often used to obtain an individual's online banking details, money can be stolen by an unauthorised person logging into the individual's account, then transferring money to another account. Stolen credit or debit card details may be used to create a copy of a card to get cash out of an ATM or to purchase goods and services at a shop or online.

Unauthorised persons can also use malware to steal intellectual property from a computer, such as artistic or literary works. Physical items can be stolen if a user's personal data such as an entry keycode to a warehouse is used to gain access to a building (see Figure 5.14).



Figure 5.14: Personal data such as door entry password codes stored on a computer may be stolen.

Industrial espionage

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Industrial espionage involves spying on organisations to gain an advantage in business (see Figure 5.15). Malware may be used to steal secrets from an organisation, such as product specifications, creative designs, upcoming releases, or contract information. Using this information, the unauthorised person could enable another organisation to gain a competitive advantage. This could include being able to launch or enhance products before their competitors, being the first to offer new services to customers, or having improved terms within contracts.

Industrial espionage may lead to the discovery of unethical or illegal business practices, which could be damaging for an organisation and could be used as part of a ransomware attack.



Figure 5.15: Industrial espionage can be used to gain improved terms within contracts.

Sabotage

Sabotage occurs when equipment or buildings are deliberately damaged or destroyed. Malware may be used to purposely delete or damage files, or make computers difficult to use, such as through disrupting network traffic.

Malware is increasingly used to target industrial equipment and infrastructure, such as power stations and electricity grids. Sabotage can be particularly dangerous if safety critical systems are targeted, such as those in nuclear reactors, or life support systems in hospitals.

Consequences of malware for organisations and individuals

While malware can sometimes be a nuisance, such as adware pop-ups and banners, the consequences of a malware attack can be severe:

- An organisation's operations may be badly disrupted or stopped. The organisation may be unable to serve its customers, which will lead to financial loss as well as a possible loss of business reputation. Customers may choose to buy from the organisation's competitors instead. There is also the time and cost of removing and resolving issues caused by malware. The consequences of a malware attack may only be evident in the longer term. For example, a competitor gains a business advantage when they launch products based on stolen product designs.
- Malware attacks take time and money for individuals to resolve as they may lack the expertise to fix any issues themselves. Not having access to a functioning digital device may limit an individual's ability to work and earn money, or mean that educational tasks, such as completing assignments, cannot be completed. Personal data stolen through malware and used for identity theft can be particularly difficult to resolve. The stress may cause the individual to suffer health issues.

Malware prevention software and physical prevention methods

Both anti-virus software and malware removal tools can be used to prevent malware attacks. The system can also be scanned for malware and infected files quarantined or removed. The technologies outlined above should also be used as required, including firewalls, VPNs, software updates and only authorised users allowed access to computer networks.

Malware attacks often result from a user's actions on a device, so it is important individuals are aware of appropriate behaviour when online and using digital devices. Organisations may provide training on malware prevention for employees. Users should be made aware of the dangers posed by email links and attachments, using USB memory sticks to transfer data and the need to check what is included as part of downloaded software.

Individuals and organisations should also react quickly should a malware attack occur, including disconnecting any infected computers from a network and using appropriate software to remove the malware.

Advantages and disadvantages of the methods of malware prevention

Many of the advantages and disadvantages of the methods used to prevent malware are outlined earlier in the chapter. Table 5.2 identifies additional advantages and disadvantages.

Method	Advantages	Disadvantages
Regularly update software and operating system.	Patches known bugs and security weaknesses used by malware. Can take place automatically, without user intervention.	Users may be unable to use the machine while the update takes place. Updates might unexpectedly alter other parts of the software or operating system.
Limit user access to a computer, including downloading software and using USB memory sticks.	Reduces the likelihood of a device being infected through downloading and / or installing malware. Malware may only be able to access data stored on a user's account, rather than the whole computer or network.	Initial setup can be time consuming, with a computer's administrator required to install software or make changes to the computer. Users can become frustrated about not being able to use the computer how they want.

Table 5.2: Additional advantages and disadvantages of malware prevention methods.